THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 11 & 12: **Keeping Time with the Skies, How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Leap years are skipped in century years like 1700, 1800 and 1900 because: 1

(a) Earth takes slightly less than 365¼ days **(b)** Moon changes
(c) Sun slows **(d)** Earth takes more than 365¼ days

2. The Indian National Calendar year begins on: 1

(a) 1 April **(b)** 21 June
(c) 22 March (21 in leap years) **(d)** 1 January

3. Festivals based on the Moon's phases, such as Eid-ul-fitr, move through the Gregorian year because: 1

(a) lunar year is shorter than the solar year
(b) Earth stops
(c) lunar year is longer
(d) Sun stops

4. Which is an Indian space observatory satellite? 1

(a) Hubble **(b)** AstroSat **(c)** Sputnik **(d)** Voyager

5. Vikram Sarabhai Space Centre is located in: 1

(a) Hyderabad **(b)** Sriharikota
(c) Thiruvananthapuram **(d)** Bengaluru

6. If producers decrease in an ecosystem, the number of herbivores will probably: 1

(a) double **(b)** increase **(c)** decrease **(d)** stay constant

7. A tapeworm in the gut of a human is an example of: 1

(a) predation **(b)** commensalism **(c)** parasitism **(d)** mutualism

8. Which practice helps soil health? 1

(a) burning crop residue **(b)** excess pesticides
(c) organic manure **(d)** monoculture

9. The Sundarbans were declared a UNESCO World Heritage Site in: 1

(a) 1987 **(b)** 1900 **(c)** 1947 **(d)** 2000

10. Bandipur National Park is in: 1

(a) Karnataka **(b)** Punjab **(c)** Gujarat **(d)** Assam

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What are equinoxes and solstices? 2

12. Mention two uses of artificial satellites. 2

13. Mention two ecosystem services. 2

14. Mention two human activities that threaten ecosystems. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is the difference between rotation and revolution of Earth? How do they define day and year? 3

16. What is the role of abiotic components in an ecosystem? 3

17. Describe the activity of observing the Moon at the same time on successive days. 3

OR

Why do the months in the Gregorian calendar have 30 or 31 days? 3

18. Explain with an example how an ecosystem can be both natural and human-made. 3

OR

Explain why the number of organisms decreases at higher trophic levels. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. A lunar year has 354 days and a solar year has 365¼ days. Calculate the difference. How does an Adhika Maasa help? 5

OR

Describe a night sky observation project to record the Moon's phases for a month. 5

20. A pond has algae, small fish, big fish and bacteria. 5

(i) Make a food chain.
(ii) Name the producer and consumers.
(iii) Role of bacteria.
(iv) Effect of removal of small fish.

OR

Describe how a change in one component, such as a drought, affects the entire ecosystem. 5

END OF QUESTION PAPER — Total: 40 Marks

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